WORKSHEET 1

Student Name: Analava Bera **UID: 23BCS13611**
Branch: BE-CSE **Section/Group: Krg_3A**
Semester: 6th **Subject Name: System Design**
Subject Code: 23CSH-314

Aim: To design a URL Shortner System

Steps:

1. **Functional Requirements:**

   Create a short URL from a Long URL.

   --Optional: Support custom URL

   --Support expiration date: Default+ custom expiration

   User should get redirected to the original URL from short URL.

2. **Non-Functional Requirements:**

   Low Latency: response in least amount of time: (On URL-creation, on URL Redirect) - 200 ms.

   Scalability: 100M daily active user & application should be able to short 1B URL.

   Unique: Shorten URL should be Unique.

   Availability: 24 x 7 available.

   CAP: Tradeoff btw Consistency & Availability: We need high availability here as per req.
   Availability >> Consistency
   We will achieve consistency having some delay - This system is called Eventual system.

3. **API Design:**

- POST Endpoint: /v1/url -> short-URL

Since this is POST request, we have to pass the body

{
longURL,
customURL ? (optional),
expirationDate}

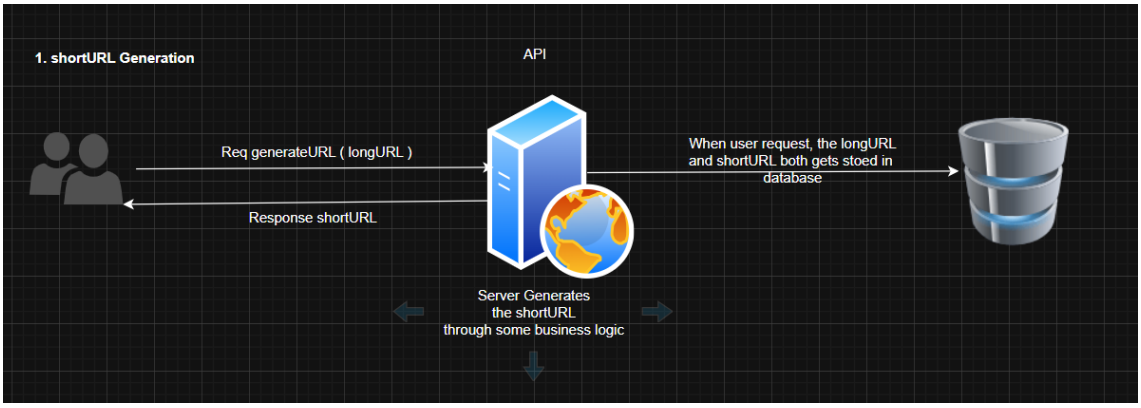
Now we have retrive the longURL also, we neet GET Request
- GET: /v1/url/{short-url} -> will redirect me to longURL.

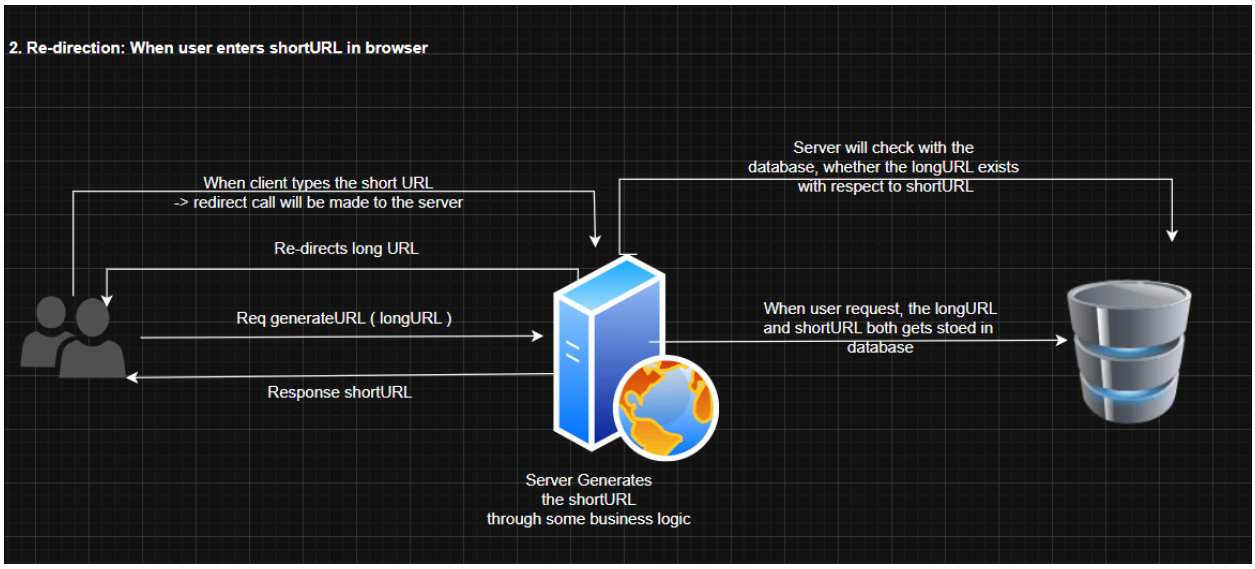
4. **High-Level Design:**

Now According to the functional requirement of the system, we can identify that :

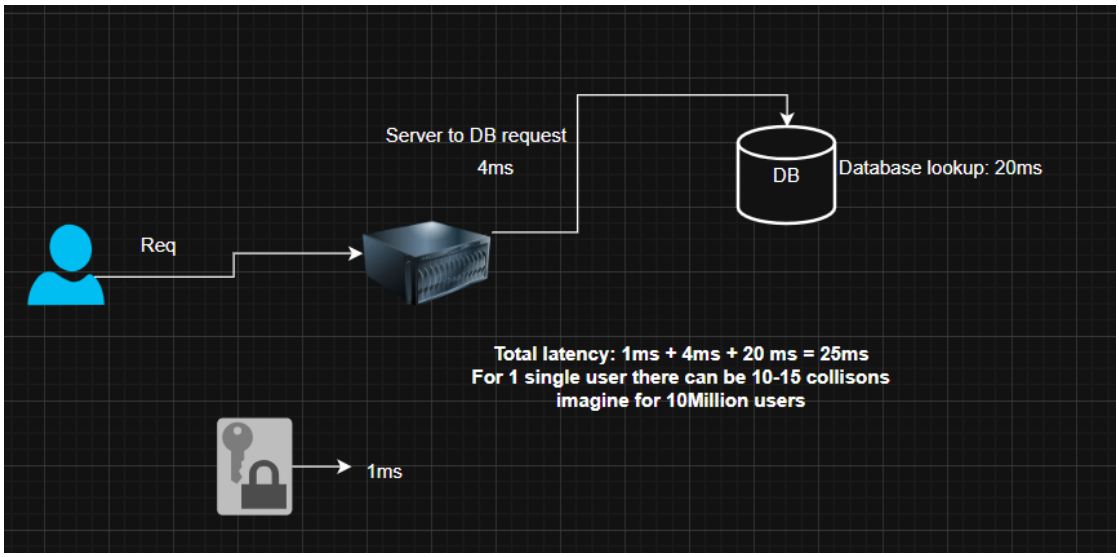
There will be a client who is requesting, then there will a server upon which computation will be going on, and lastly there will be an database in which storage will be done.

5. **Low-Level Design:**

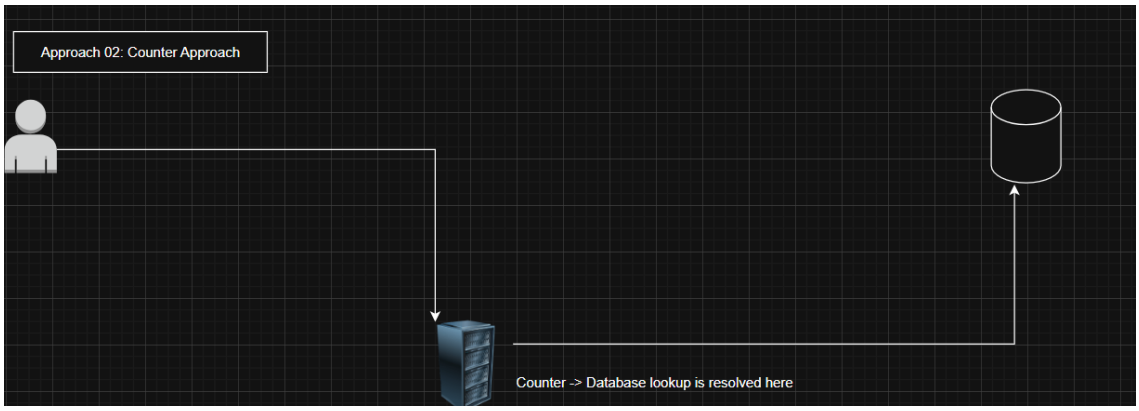




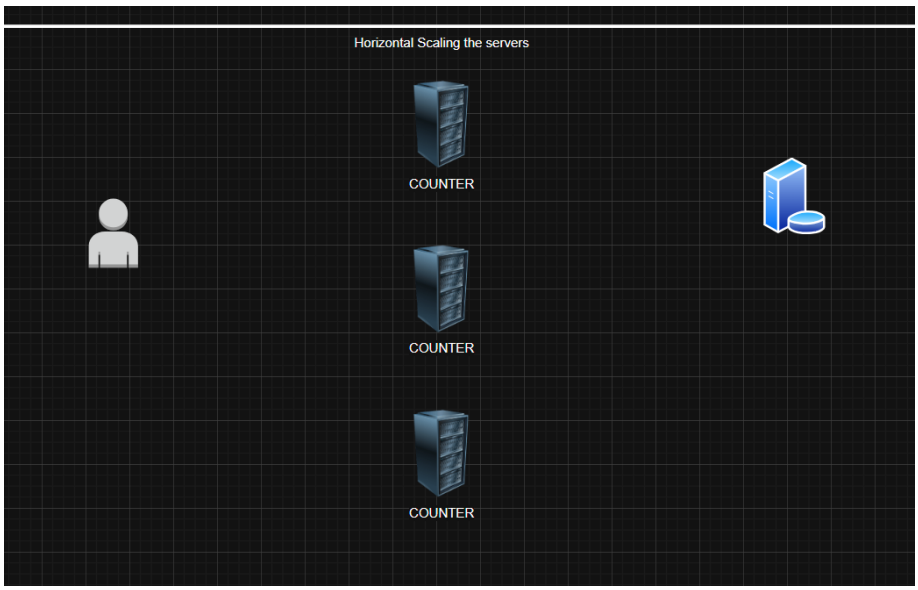
Approach 01: Encryption

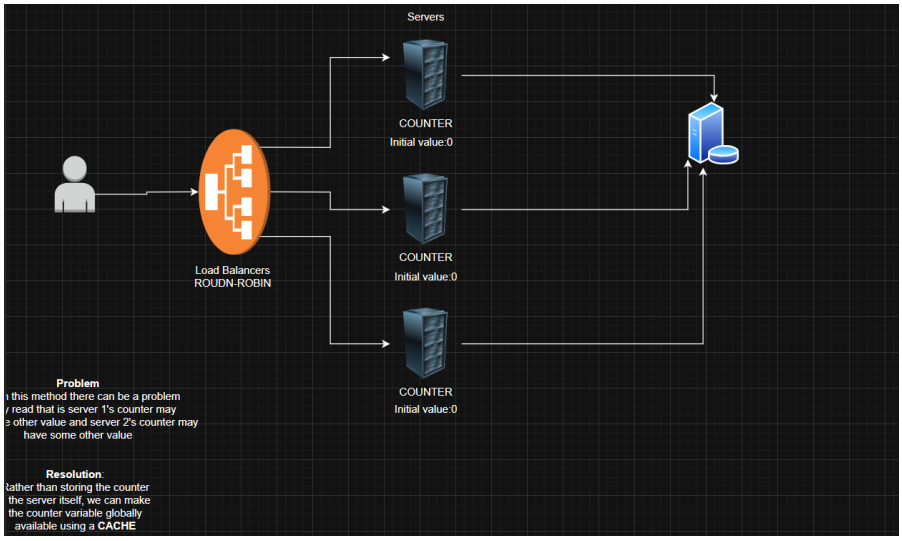


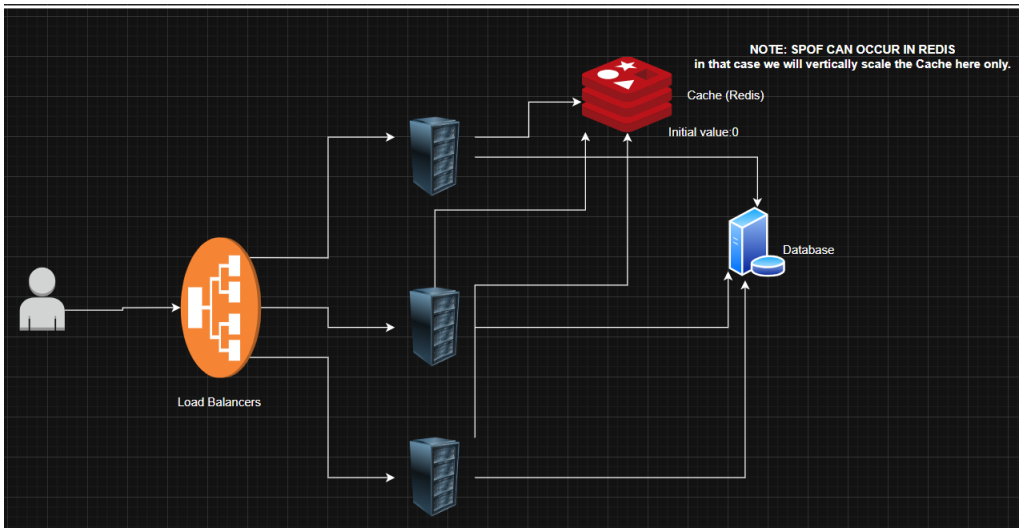
Approach 02: Counter Approach



Scaling:


Horizontal Scaling the servers
COUNTER
COUNTER
COUNTER


Servers
COUNTER
Initial value:0
COUNTER
Initial value:0
COUNTER
Initial value:0
Load Balancers
ROUDN-ROBIN
Problem
this method there can be a problem
read that is server 1's counter may
other value and server 2's counter may
have some other value
Resolution:
ather than storing the counter
the server itself, we can make
the counter variable globally
available using a CACHE


NOTE: SPOF CAN OCCUR IN REDIS
in that case we will vertically scale the Cache here only.
Cache (Redis)
Initial value:0
Database
Load Balancers